Aaron Kumar

3× International Award Winner | McMaster Rocketry Team | Mechanical Engineering Student
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Experience

Lean Manufacturing Intern May – Aug. 2026
Canam Group *Mississauga, ON*

- Led an A3 consumables cost reduction project, delivering $22,790.65 in savings (79.4% total reduction) through root-cause analysis of consumption drivers and standardized usage controls.
- Proposed an efficiency upgrade for Roll Former #14, projected to increase line uptime by up to 4.7%.
- Raised 5S audit scores across 3 production areas by an average of 4.3%.
- Designed and deployed a portable metrology jig that cut a 3-week frame inspection cycle to under one minute (see Projects).

Propulsion Team Member Sep. 2025 – Present
McMaster Rocketry Team *Hamilton, ON*

- Contribute to the design and development of McMaster University's first hybrid rocket engine as part of the propulsion sub-team.
- Support mechanical design, subsystem development, and hardware testing across propulsion components, contributing to systems integration and iterative design refinement.

Initiative Contributor Sep. 2023 – Jun. 2025
Peel Engineering and Math Learning Initiative *Ontario, Canada*

- Supported outreach efforts expanding student access to engineering, math, and applied technical learning.

Teaching Assistant / Instructor Sep. 2022 – Jun. 2025
Spirit of Math *Ontario, Canada*

- Delivered advanced mathematics instruction in an enrichment-focused, competitive environment, building student problem-solving and logical reasoning skills.

Projects

Portable Metrology Jig | *Mech. Design, Metrology* 2026

- Designed and built a portable inspection fixture functioning as a self-contained metrology lab, scanning surface defects and measuring feature-to-feature distances across a full structural frame on the shop floor.
- Replaced a 3-week scan-and-transport workflow with a sub-one-minute in-place inspection; owned concept CAD, prototyping, and on-site validation.

Robotic End Effector | *CAD, Robotics, Python* 2025

- Designed and 3D-printed a gear-driven end effector for a Q-arm warehouse fulfillment system (ENG1P13); served as Subject Matter Expert, leading all CAD iterations and exploded drawings.
- Redesigned final gripper to improve torque output; implemented Python `customer_summary()` function.

Motorized Storage Device | *Mech. Design, Embedded Systems* 2026

- Built an Arduino-controlled rack-and-pinion motorized lift for a powerchair user (ENG1P13); led all CAD and ∼80% of electronics. Used BambuLab P1S for rapid same-day prototyping.

ASTRAEUS II: Eden Garden Module | *Space Systems, CAD* 2025

- Co-developed award-winning space agriculture habitat; led structural design and systems integration of a sustainable extraterrestrial module supporting up to 592,391 inhabitants. Presented at NSS international competition (1st place, 2024 & 2025).

Summary

Second-year Mechanical Engineering student at McMaster University with strong interests in aerospace, propulsion, and mechanical system design. Member of the McMaster Rocketry Team, working on McMaster's first hybrid rocket engine. 3× international award winner with experience in space systems design, CAD, sustainable engineering, and multidisciplinary project leadership. Completed a Lean Manufacturing internship with Canam Group's Operational Excellence team.

Education

McMaster University
B.Eng., Mechanical Engineering
Hamilton, ON | Sep. 2025 – Present

Central Peel Secondary School
Ontario Secondary School Diploma
Brampton, ON | Sep. 2021 – Jun. 2025

Awards & Leadership

- **1st Place** – Live in a Healthy Space Design Competition, NSS *(2025)*
- **1st Place** – Live in a Healthy Space Design Competition, NSS *(2024)*
- **Honourable Mention** – Gerard K. O'Neill Space Settlement Contest, NSS *(2024)*
- Recognized by Peel District School Board & City of Brampton for STEM leadership
- Founded **Advancing Future Innovators (AFI)**, delivering hands-on STEAM workshops to students across Ontario

Technical Skills

CAD: Autodesk Inventor, Onshape, Shapr3D

Engineering: Propulsion Systems, Aerospace Design, Mechanical Prototyping, Space Systems, Lean Manufacturing, 5S, A3 Problem Solving

Technical: Metrology & Inspection, Hardware Testing, Research, Technical Presentations, Team Collaboration

Programming: Python

Projects: Hybrid Rocket Engines, Space Habitat Design, Hydrogen Generation, Experimental Propulsion